

Cross-Temporal Coherence: Minimum Structural Requirements

Formal Constraint Specification from Navigational Cybernetics 2.5

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Glossary of Key Terms

Term	Meaning in NC2.5
τ (internal time)	Remaining structural viability budget. Monotonically decreasing. Not clock time.
Φ (structural burden)	Total accumulated irreversible structural load. Monotonically increasing.
Admissibility	A non-causal structural predicate that constrains realization without providing gradient signal or optimization objective.
Coherence	Internal organizational alignment across perception, interpretation, action, and temporal coordination. Not a performance metric.
Effect-class	Equivalence class of structural consequences that an action induces on long-horizon continuation.
Phase debt	Accumulated unresolved phase-related residuals and regime-transition load.
Non-reconstructibility (NR)	Formal bounds (NR- ϵ , NR-LR) preventing admissibility geometry from being inferred from gate emissions.
Regime	An organizational mode of the system. Transitions between regimes may be irreversible.
Gate (ingress/egress)	Structural boundary that filters operations. Ingress = entry. Egress = irreversible commitment.
Runtime- τ vs audit- τ	NC2.5 allows τ to be deployed as a runtime variable or an audit-only observable. This spec does not prescribe which; the choice is an implementation decision for the persistence layer.

1. Minimum Structural Requirements for Cross-Temporal Coherence

Cross-temporal coherence is the property that an adaptive system's internal organization remains structurally continuous across time — not merely behaviorally consistent, but organizationally invariant under admissible evolution.

The following requirements are derived from NC2.5 axioms and theorems. Any semantic persistence layer that claims to preserve cross-temporal coherence must satisfy all of the following:

R1. Identity is carried by coherence, not by behavior (Axiom 20)

Identity continuity is carried by coherence of internal organization rather than by persistence of observable behavior. Behavioral similarity across time does not guarantee preservation of identity.

Constraint: A persistence layer must track internal organizational invariants, not behavioral outputs. Two states producing identical behavior may have different identity status.

R2. Structural properties are long-horizon and non-local (Axiom 14)

Structural viability, identity continuity, and coherence are global temporal properties. They cannot be inferred from instantaneous state, action quality, local error, reward signals, or short-horizon performance metrics.

Constraint: A persistence layer cannot determine coherence from snapshots. It must maintain and evaluate trajectory-level properties across the full operational history.

R3. Coherent closure has finite validity horizon (Theorem 2)

In any long-horizon adaptive system operating under non-zero directed interaction, no coherently closed structure can remain admissible indefinitely. Every coherently closed commitment has a finite structural validity horizon measured in remaining internal time.

Constraint: No persisted state or commitment may be treated as permanently valid. Its validity horizon must be indexed to internal time and accumulated structural burden, not to wall-clock time or arbitrary TTL. (The specific threshold or expiration mechanism is a lab-level implementation decision, not an NC2.5 derivation.)

R4. Structural burden accumulates independently of trajectory correctness (Axiom 18)

Structural burden may accumulate even when trajectories remain admissible, errors remain bounded, and performance metrics remain stable. Trajectory correctness does not bound structural consumption.

Constraint: A persistence layer must account for burden accumulation even when the system reports no errors. Correct behavior does not imply structural health.

R5. Regime transitions may be irreversible (Axiom 21)

There exist regime transitions after which recovery of admissible behavior does not imply recovery of prior identity. Such transitions constitute irreversible structural events even in the absence of observable failure.

Constraint: A persistence layer must detect and record irreversible regime transitions. Recovery of behavior after such a transition does not mean identity was preserved. This is a one-way gate.

R6. Context is not meaning (Axiom 23)

Context is a transient interpretive structure. Meaning is context that has preserved structural admissibility across internal time. A persistence layer must not treat context persistence as meaning preservation.

Constraint: Persisted contexts must be evaluated for structural admissibility before they are treated as stable meaning. Premature closure of context into meaning (Axiom 26 — Context Capture) is a structural failure mode.

R7. Long-horizon viability requires a non-causal observational dimension (Axiom 22; supported by Theorem 36, Corollary 37.1)

Long-horizon viability and identity continuity cannot be reliably assessed from within the causal decision-making process of an adaptive system. Any architecture lacking a non-causal observational dimension is structurally incapable of detecting silent long-horizon degradation.

Constraint: The persistence layer must operate as a non-causal observer. If it is folded into the causal decision loop — as a reward signal, optimization input, or policy parameter — it loses the ability to detect the degradation it exists to measure. This is the foundational reason for constraints C1–C4 in Section 2.

2. Interface Specification with Admissibility and Gate Structure

The semantic persistence layer operates in the space between the ingress gate and the egress gate. It does not authorize actions. It does not enforce governance. It provides structural state to the admissibility evaluation without participating in it.

2.1. Inputs from Admissibility Layer

Input	Source	Description
Current τ value	Internal time budget	Remaining structural viability. Monotonically decreasing. (This spec assumes runtime- τ deployment. Under audit- τ deployment, this input is replaced by an external audit signal.)
Accumulated Φ	Burden functional	Total irreversible structural load. Monotonically increasing.
Active regime identifier	Regime classification	Current operational regime of the system.
Phase history h	Phase accumulator	Sequence of regime transitions and associated costs.
Candidate operation u	Action pipeline	The operation under evaluation. Required to determine which continuation is being assessed.
Structural horizon H	Horizon estimator	The evaluation horizon over which admissibility is being assessed.

2.2. Outputs to Admissibility Layer

Output	Consumer	Description
Coherence indicator	Gate evaluation	Threshold-based measure of internal organizational alignment: above/below coherence threshold, not a continuous scalar. This is a lab-side witness variable, not an NC2.5 primitive. Not a reward signal. Not an optimization target.
Contraction indicator	Viability assessment	Binary signal: continuation-space contracting or not.
Irreversibility flag	Regime tracker	Signal that an irreversible regime transition has occurred since last evaluation.
Burden-rate estimate	τ -budget monitor	Coarsened rate category (low / elevated / critical) of structural burden accumulation. Audit-only; not visible to acting agent's optimization surface.

2.3. Architectural Constraints on the Interface

C1. No feedback loop. The persistence layer provides state to the admissibility evaluation. The admissibility evaluation does not modify the persistence layer's internal dynamics. This prevents admissibility from collapsing into an optimization target for the persistence mechanism.

C2. No gradient signal; gate-only visibility. Outputs from the persistence layer are categorical or threshold-based. They do not provide continuous gradient that could be used to optimize toward or away from admissibility boundaries. Furthermore, all persistence outputs are restricted to gate-only or audit-only consumption. They must not be visible to the acting agent's optimization process. This prevents the emissions from becoming geometry signals that could be exploited to reconstruct admissibility boundaries, violating non-reconstructibility constraints (NR- ϵ , NR-LR).

C3. Non-causal observation. The persistence layer observes structural evolution but does not intervene in it. It is a witness, not an actor. This is supported by Axiom 15 (causal observation distorts long-horizon properties), Axiom 29 (non-causal admissibility), and the gate bifurcation results (Theorems 36–37) establishing that non-causal witnessing is structurally necessary for silent-degradation detection.

C4. Gate independence. The persistence layer does not determine what passes through ingress or egress gates. It provides structural context that gates may use, but the gate decision is architecturally separate.

2.4. Inherited Constraints from NC2.5

The following NC2.5 formal devices are inherited by the persistence interface and must not be violated by any implementation:

Effect-class discipline. Admissibility is evaluated over structural effect-classes $E(S, u, H, ctx)$, not over actions directly. The persistence layer provides partial structural observables (S, H, ctx) that feed into effect-class construction, but does not perform final admissibility classification. Adequacy conditions EC-1 through EC-4 apply to any component that feeds into effect-class construction.

Authorization-before-evaluation (Axiom 31; Theorems 40, 41). No operation reaches admissibility evaluation without prior authorization. The persistence layer does not bypass this sequence.

Gate-output minimality and non-reconstructibility (Axioms 51, 58, 59; Theorem 69). Gate emissions must be minimal. The persistence layer's outputs must not enable reconstruction of the admissibility boundary geometry. This is what makes admissibility non-gradient and non-optimizable — not merely the label "no gradient signal", but the formal NR- ϵ / NR-LR / NR-Q bounds on information leakage through emissions.

Zero executability of observation (Axiom 32). Persistence-layer observations have zero executability: they cannot be converted into action selection signal by the acting agent.

3. Formal Conditions Under Which Persistence Failure Constitutes Viability Loss

3.1. Definition of Persistence Failure

Persistence failure occurs when the semantic persistence layer can no longer sustain the organizational conditions required by R1–R7 — that is, when internal coherence, burden tracking, irreversibility detection, or non-causal observation capacity degrades below operational viability. This is not an error. It is a structural boundary condition.

3.2. Persistence Failure $\rightarrow$ Viability Loss (Theorem 2 + Axiom 27)

Condition 1 (lab convention, not NC2.5 derivation): If coherence indicator drops below a structural threshold θ_{coh} and does not recover within a bounded horizon Δt , then the system has exited the cross-temporal coherence class for the purposes of this test. Note: θ_{coh} and Δt are lab-level boundary conventions chosen for the current stress test, not formally derived from NC2.5. What NC2.5 provides is Theorem 2 (finite validity horizon under non-zero drift) — the specific thresholds operationalize that theorem for testing.

Condition 2: If sustained non-closure occurs with monotonically accumulating phase debt and a lower-bounded τ -depletion rule holds, then internal time is exhausted in finite time (Theorem 57). Note: Theorem 57 requires specific antecedents — sustained non-closure, monotone phase debt, and $f(\kappa, D, C) \geq \alpha D$ for $\alpha > 0$. This condition applies only when those premises are satisfied, not as a general race between burden rate and maintenance rate.

Condition 3: If an irreversible regime transition occurs (R5) and the persistence layer cannot establish coherence in the new regime, the system has undergone identity discontinuity. By Axiom 21, recovery of admissible behavior after such a transition does not imply recovery of prior identity. The persistence layer must signal this as a terminal structural event for the prior identity.

3.3. Falsification Criteria

The following observations would falsify the persistence-viability coupling:

1. A system maintains cross-temporal coherence indefinitely under sustained coupled load and non-zero interaction pressure with zero structural burden accumulation $\rightarrow$ violates Axiom 61 / Theorem 65 (burden-consumption under sustained coupled load), and by extension the general Axiom 27 line.
2. A system recovers prior identity after irreversible regime transition by behavior recovery alone $\rightarrow$ violates Axiom 21.
3. A system maintains structural viability indefinitely while operating under non-zero directed interaction with non-zero drift accumulation and no renewal or re-closure of coherent structures $\rightarrow$ violates Theorem 2 under its stated premises (finite validity horizon requires non-zero drift).

Any of these observations indicates that the system does not belong to the architectural class defined by NC2.5.

4. Application to Bounded-Agency Stress Testing

This specification is scoped to the current phase of the Genesis AiX Interop Lab bounded-agency problem. The following mapping connects the formal requirements to the lab's test framing:

What the lab is testing: whether a multi-participant architecture can maintain coherent governance under sustained operational pressure without silent structural degradation.

Where cross-temporal coherence enters: the bounded-agency problem assumes that participants maintain identity and semantic persistence across sessions, phases, and regime transitions. Without a formal constraint on what persistence must satisfy, this assumption is untestable — the gap Ali's SPF layer was designed to fill.

What this spec provides as interim coverage:

- R1–R7 define the minimum bar: any persistence mechanism used in the stress test must satisfy these requirements, or the test results are structurally uninterpretable with respect to identity continuity.
- The interface specification (Section 2) ensures the persistence layer does not contaminate the admissibility evaluation — preventing the bounded-agency test from silently collapsing its own governance structure.
- The failure conditions (Section 3) make persistence failure **explicit and detectable** in the test framing, rather than silently assumed away — which is exactly what Richard's circulation requested.

What this spec does not provide: operational implementation, specific data structures, session serialization formats, or runtime APIs. These are implementation decisions that belong to whoever builds the persistence layer — whether that is Ali's SPF formalism upon his return, or an interim engineering solution.

Theorem 59 (Double-Budget Structural Closure) applies here under its stated premises: runtime- τ budget with $f \geq \beta C$, a bounded revision budget V_G with each revision costing at least $\alpha > 0$, and NR

constraints preventing admissibility reconstruction from gate emissions. Under those premises, the persistence layer cannot escape structural constraints through infinite meta-revision or infinite regime switching. A persistence mechanism that relies on unlimited self-repair does not satisfy these premises and would invalidate the bounded-agency test.

Theorem 67 (Coordination-Enforcement Requirement) is a missing dependency that this note does not discharge. Because the lab is multi-participant and subject to turnover, truncation, and model-update perturbations, the coordination constraints identified by Theorem 67 apply. This spec covers the formal constraint layer for persistence; it does not cover the multi-agent enforcement layer that Theorem 67 requires.

Summary

Deliverable	NC2.5 Source	Key Constraint
R1–R7: Minimum structural requirements	Axioms 14, 18, 20, 21, 22, 23, 26; Theorem 2	Coherence = organizational invariants, not behavior. Finite validity. Burden independent of correctness. Irreversibility. Context $\neq$ meaning. Non-causal observation required.
Interface specification	Axioms 15, 22, 32; Theorem 69; NR constraints	Non-causal observation. No feedback. No gradient. Gate independence. Inputs: τ , Φ , regime, history, candidate operation u , horizon H . Outputs: threshold-based coherence, binary contraction, irreversibility flag, coarsened burden-rate (audit-only). Inherited: effect-class discipline EC-1..EC-4, authorization-before-evaluation, non-reconstructibility.
Persistence failure = viability loss	Theorems 2, 57; Axioms 21, 27	Condition 1: coherence below threshold without recovery (lab convention, not NC2.5 derivation). Condition 2: finite exhaustion under Theorem 57 premises (sustained non-closure, monotone phase debt, $f \geq \alpha D$). Condition 3: irreversible regime transition without new coherence = identity discontinuity (Axiom 21).
Bounded-agency application	Theorem 59 (under premises); Theorem 67 (missing dependency)	Persistence gap is testable, not assumed. Double-budget prevents infinite self-repair under stated premises. Coordination-enforcement (Theorem 67) not discharged by this spec.

This document is Class 1 material: private reference, formal constraint specification. Not operational implementation. It does not replace Ali Mofradi's SPF formalism. It provides the structural substrate that any cross-temporal persistence layer must satisfy within the NC2.5 architectural class.

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